

AIM

To write a Python program to implement the *A Search Algorithm** for finding the shortest path between two nodes.

OBJECTIVE

- To understand the A* Search Algorithm.
- To find the shortest path between the source and destination.
- To use heuristic values for efficient searching.
- To demonstrate informed search in Artificial Intelligence.

ALGORITHM

Step 1: Define the graph and heuristic values.

Step 2: Insert the start node into the open list.

Step 3: Calculate the evaluation function:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = Cost from the start node.
- $h(n)$ = Estimated cost to reach the goal.

Step 4: Select the node with the lowest $f(n)$.

Step 5: If the selected node is the goal node, display the shortest path.

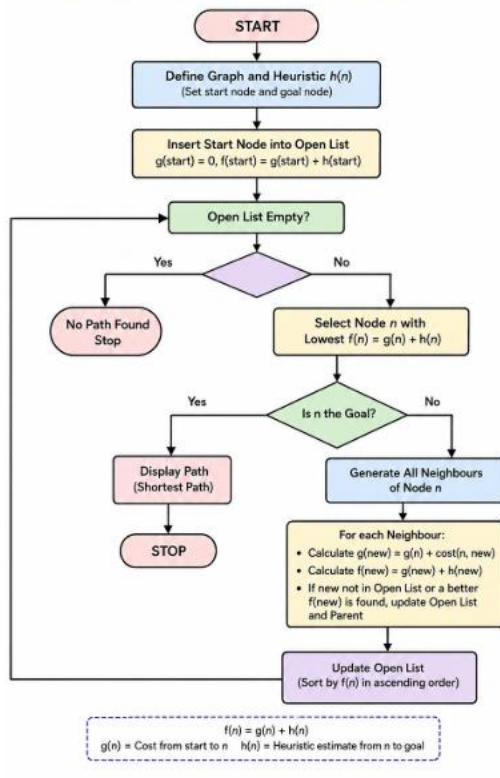
Step 6: Otherwise, generate its neighboring nodes.

Step 7: Update their cost and heuristic values.

Step 8: Repeat until the goal node is reached.

FLOWCHART

A* SEARCH ALGORITHM – FLOWCHART



PYTHON CODE

```
graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 3), ('E', 6)],
    'C': [('F', 5)],
    'D': [],
    'E': [('G', 2)],
    'F': [('G', 2)],
    'G': []
}
```

```
heuristic = {
    'A': 7,
    'B': 6,
    'C': 4,
    'D': 5,
    'E': 2,
    'F': 1,
    'G': 0
}
```

```
open_list = [('A', 0)]
visited = set()
```

```
while open_list:

    open_list.sort(key=lambda x: x[1] + heuristic[x[0]])

    node, cost = open_list.pop(0)

    if node in visited:
        continue

    print(node, end=" ")

    visited.add(node)

    if node == 'G':
        print("\nGoal Reached")
        break

    for neighbour, weight in graph[node]:
        if neighbour not in visited:
            open_list.append((neighbour, cost + weight))
```

SAMPLE OUTPUT

A B C E F G

Goal Reached

CONCLUSION

The *A Search Algorithm** was implemented successfully. It uses both the actual cost and heuristic cost to efficiently find the shortest path to the goal node. Compared to uninformed search algorithms, A* reduces unnecessary node exploration and provides an optimal solution when the heuristic is admissible.